Equations of Planes: Scalar Product Equations

Questions 1–11

Question 1. Write down the equation of the plane passing through the point $(-1, 8, 2)$ and perpendicular to the vector $-2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, giving your answer in the form $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$.

Question 2. It is given that the planes Π_1 and Π_2 have equations $(\mathbf{r}-\mathbf{i})\cdot(2\mathbf{i}+\mathbf{k}) = 0$ and $(\mathbf{r}-3\mathbf{i}+2\mathbf{j})\cdot(6\mathbf{i}-\mathbf{j}+2\mathbf{k}) = 0$ respectively. Find the acute angle between Π_1 and Π_2, giving your answer in degrees to 1 decimal place.

Question 3. It is given that the plane Π has equation

$$\left(\mathbf{r} - \begin{pmatrix} 5 \\ 2 \\ -3 \end{pmatrix}\right) \cdot \begin{pmatrix} -6 \\ 5 \\ -1 \end{pmatrix} = 0.$$

Show that the point with coordinates $(3, \frac{8}{5}, 7)$ lies on the plane.

Question 4. It is given that the plane Π has equation

$$\mathbf{r} \cdot \begin{pmatrix} -9 \\ 8 \\ -16 \end{pmatrix} = 6.$$

Show that the point $(2, -5, -4)$ lies on the plane Π.

Question 5. Write the equation of each of these planes in cartesian form, $ax + by + cz = d$, where a, b, c and d are integers:

a) $\mathbf{r} \cdot \mathbf{i} = 2$
b) $\mathbf{r} \cdot \mathbf{j} = 0$
c) $\mathbf{r} \cdot \mathbf{k} = -3$
d) $\mathbf{r} \cdot (3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) = 9$
e) $\mathbf{r} \cdot (10\mathbf{i} - 50\mathbf{j} + 25\mathbf{k}) = 30$

Question 6. It is given that the plane Π is perpendicular to the vector $2\mathbf{i} - 3\mathbf{j} - 10\mathbf{k}$ and contains the point $(3, -6, 8)$. Find the vector equation for the plane Π in the form $\mathbf{r} \cdot \mathbf{n} = p$.

Question 7. Find the acute angle between the following pairs of planes:

a) $\Pi_1 : \mathbf{r} \cdot (-\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}) = 9, \quad \Pi_2 : \mathbf{r} \cdot (3\mathbf{i} + \mathbf{j} + 10\mathbf{k}) = 2$

b) $\Pi_1 : \mathbf{r} \cdot (-5\mathbf{i} - \mathbf{j} - 9\mathbf{k}) = 1, \quad \Pi_2 : \mathbf{r} \cdot (16\mathbf{i} + 2\mathbf{j} - 5\mathbf{k}) = 13$

Question 8. It is given that the plane Π_1 has equation $\mathbf{r} \cdot (3\mathbf{i} - 2\mathbf{j} - 10\mathbf{k})$ and the plane Π_2 has equation $\mathbf{r} \cdot \mathbf{k} = 0$. Find the acute angle between Π_1 and Π_2.

Question 9. It is given that the plane Π_1 is perpendicular to the vector $8\mathbf{i} - \mathbf{j} + \mathbf{k}$ and contains the point $(2, 3, 1)$. It is also given that the plane Π_2 is perpendicular to the vector $\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$ and contains the point $(3, 9, 2)$.

a) Find the vector equation for the plane Π_1 in the form $\mathbf{r} \cdot \mathbf{n} = p$.
b) Find the vector equation for the plane Π_2 in the form $\mathbf{r} \cdot \mathbf{n} = p$.
c) Find the exact value of the cosine of the acute angle between Π_1 and Π_2.

Question 10. A particle moves along a path relative to a fixed origin O with equation:

$$\mathbf{r} = (800t^2 + 540t - 99)\mathbf{i} + (300 - 20t)\mathbf{j} + (800t^2 + 620t - 133)\mathbf{k},$$

where t is the time in seconds and distances are measured in metres. A plane has equation $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = 0$.

a) Find the exact time when the particle hits the plane.

The particle has velocity (in ms^{-1}) given by the following equation:

$$\mathbf{v} = (1600t + 540)\mathbf{i} - 20\mathbf{j} + (1600t + 620)\mathbf{k}.$$

a) Find the direction of the particle when the particle hits the plane.
b) Find the acute angle at which the particle hits the plane, giving your answer in degrees to 1 decimal place.

Question 11. The points $A(150, 0, -200)$, $B(300, 50, -150)$ and $C(400, 150, -300)$ all lie in the plane Π.

a) Find the equation of the plane Π in the form $\mathbf{r} \cdot \mathbf{n} = p$.

b) Find the acute angle that the plane Π makes with the plane $z = 0$.